RELAY AND FUSE BLOCK REMOVAL/INSTALLATION

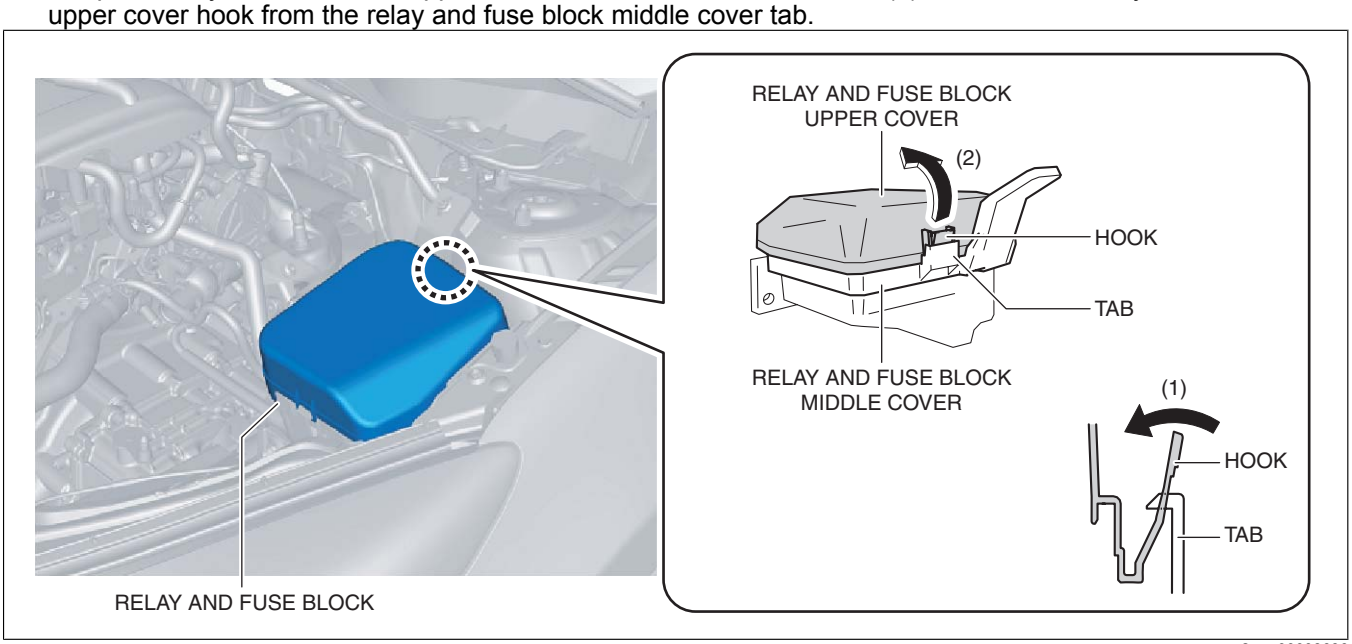
id092100498600

Note

- When replacing the relay and fuse block, it is necessary to switch terminals from the relay and fuse block terminal prior to replacement to the relay and fuse block terminal on the replacement. When switching terminals, ready the replacement relay and fuse block before pulling out the terminal of the relay and fuse block prior to replacement. This is because it is necessary to switch the terminals one at a time so that the installation positions are not mistaken.

Removal

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the air cleaner assembly. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. While pressing the relay and fuse block upper cover hook in the direction of the arrow (1) shown in the figure, lift up the relay and fuse block upper cover in the direction of the arrow (2) to detach the relay and fuse block upper cover hook from the relay and fuse block middle cover tab.



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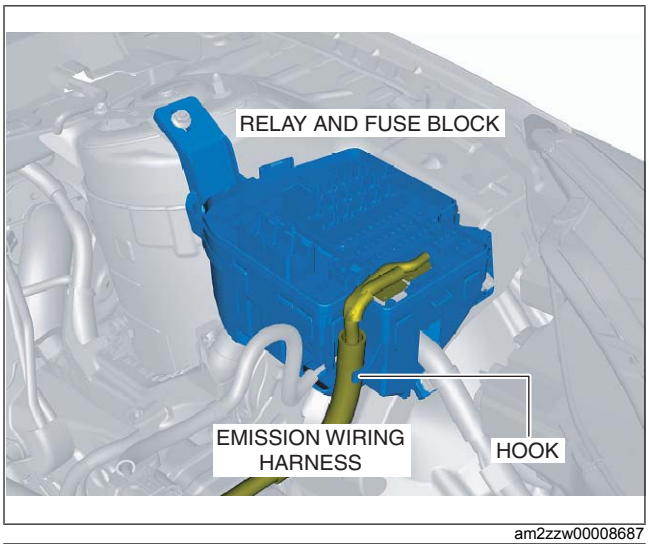
5. Remove the relay and fuse block upper cover.

Caution

- **If the relay and fuse block is removed with the relays and fuses installed, the area around the relay and fuse block may contact the relays or fuses and damage the them. Before removing the relay and fuse block, remove all the relays and fuses.**

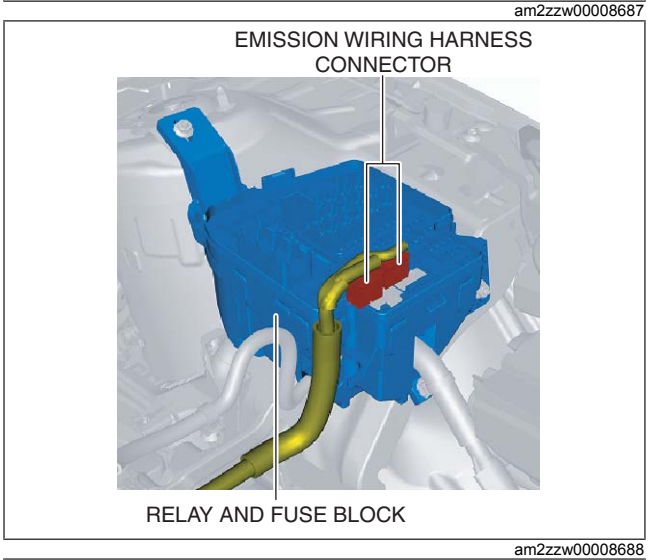
6. Remove all the relays and fuses.

7. Remove the emission wiring harness from the relay and fuse block hook.



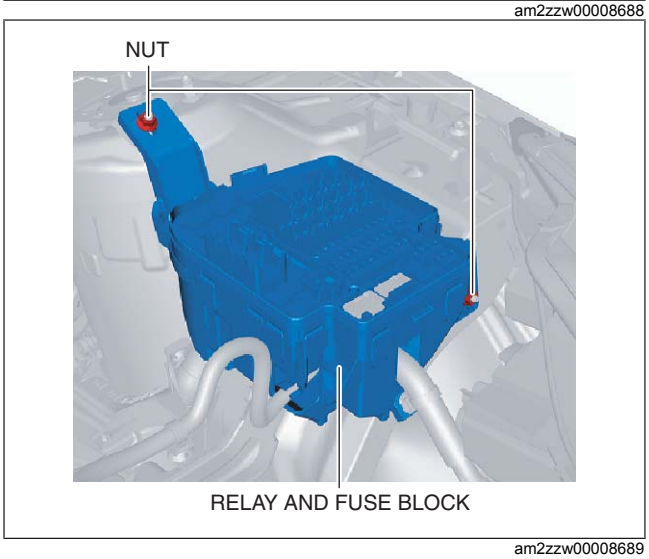
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8. Disconnect the emission wiring harness connectors.



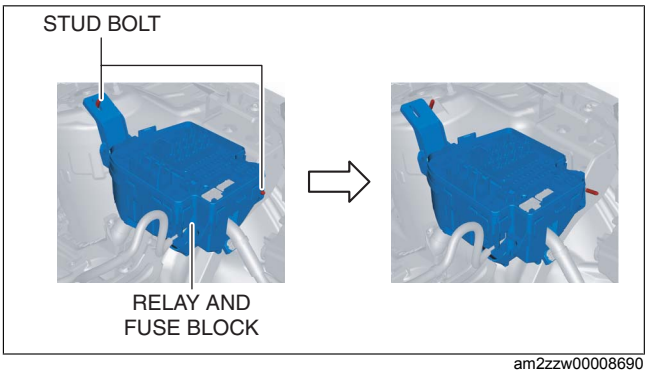
am2zzw00008688

9. Remove the nuts.



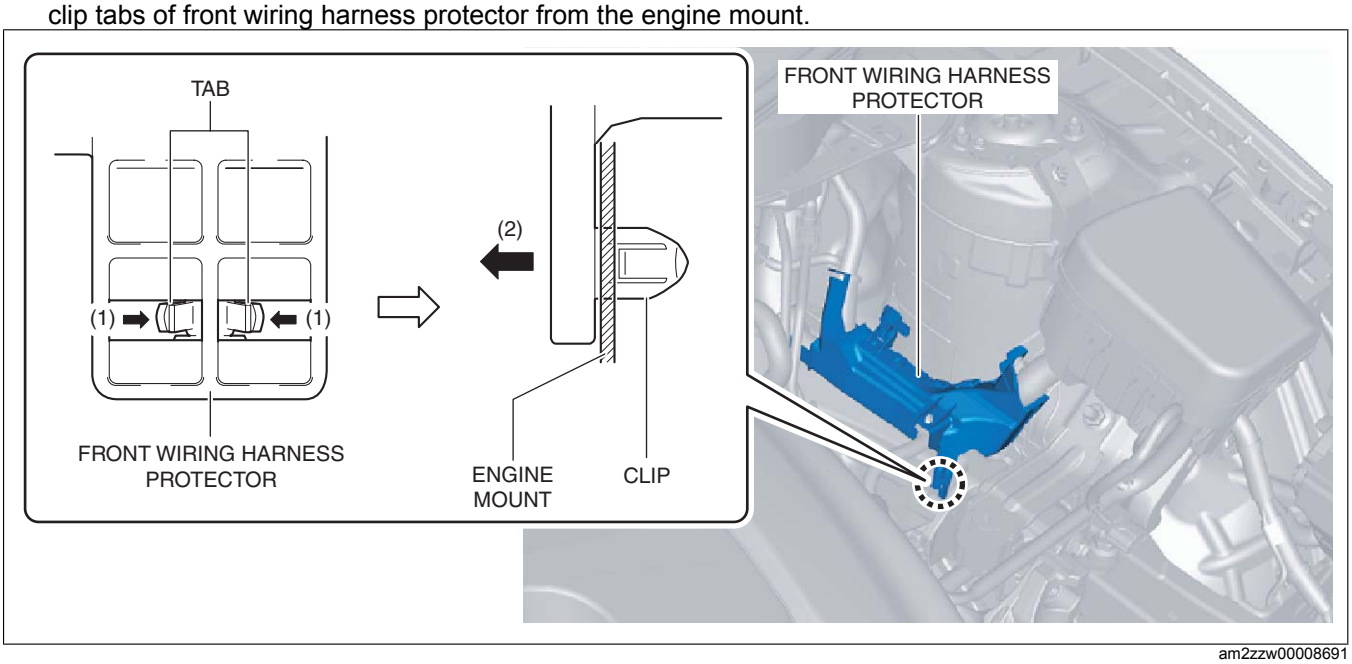
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10. Pull out the relay and fuse block from the stud bolts.



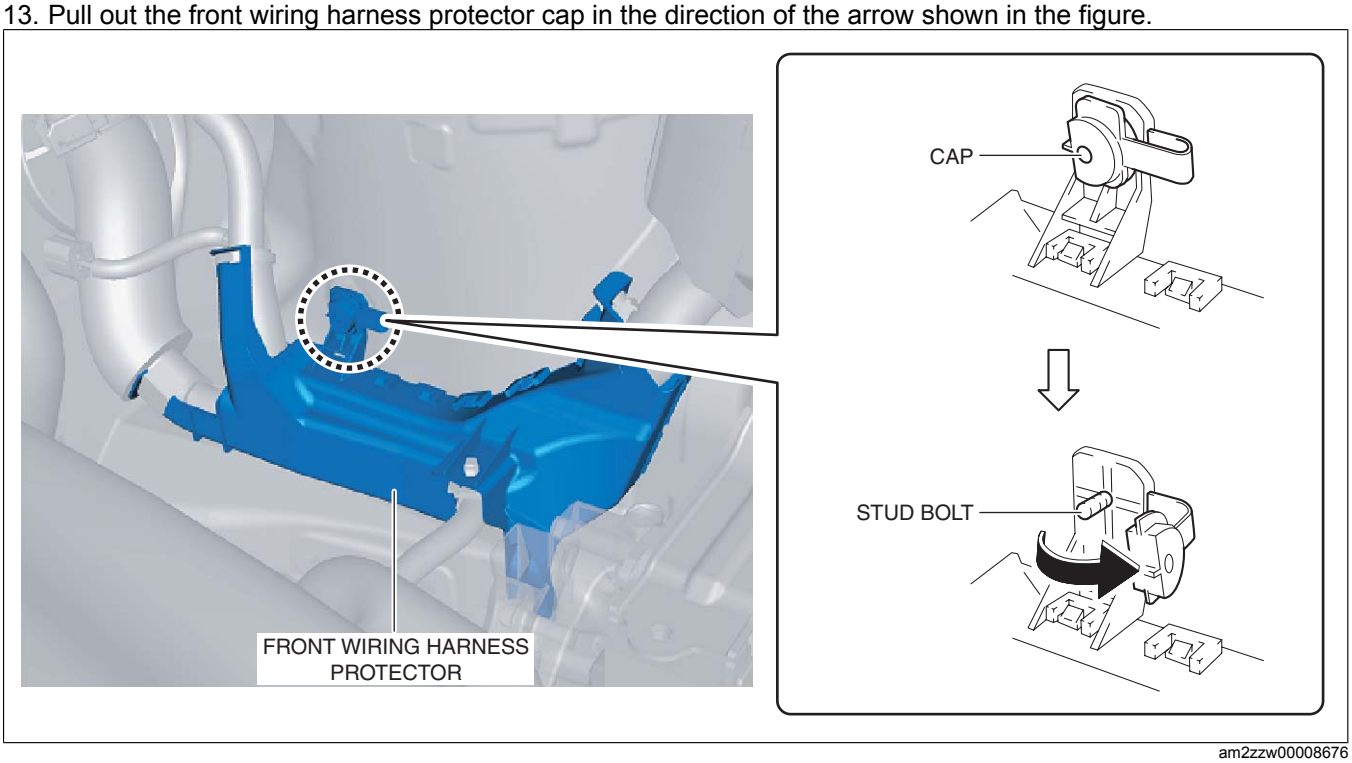
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11. While pressing the clip tabs of front wiring harness protector in the direction of the arrows (1) shown in the figure, pull the clip of front wiring harness protector in the direction of the arrow (2) shown in the figure to detach the clip tabs of front wiring harness protector from the engine mount.



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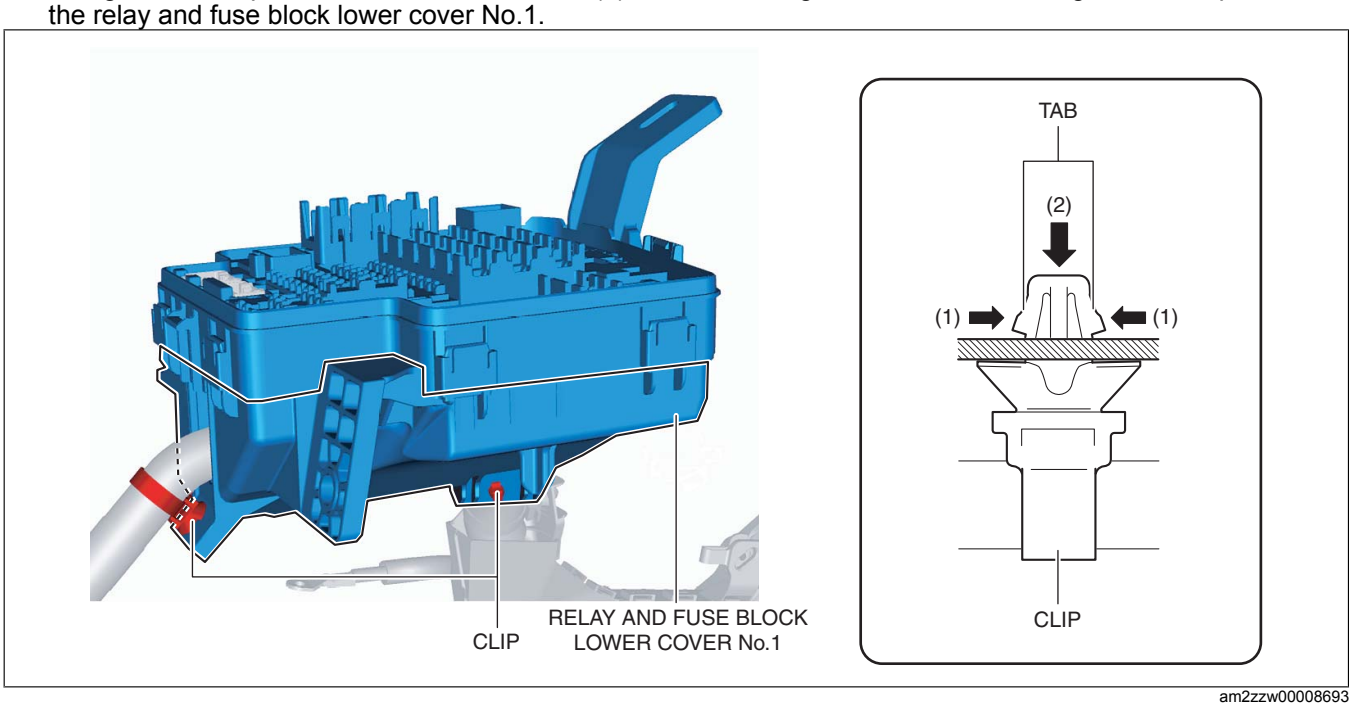
12. Pull out the front wiring harness protector clip from the engine mount.
13. Pull out the front wiring harness protector cap in the direction of the arrow shown in the figure.



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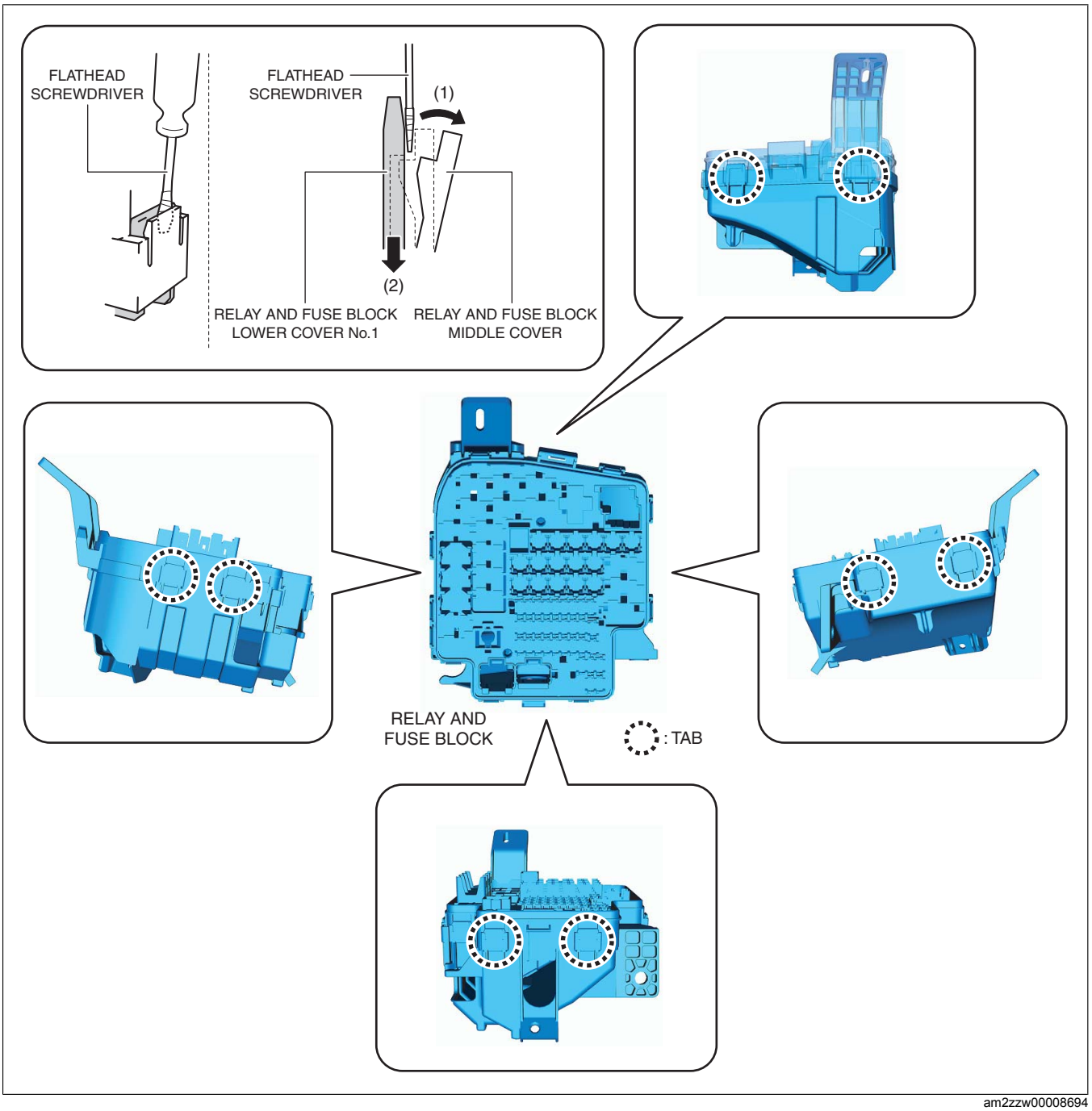
14. Pull out the front wiring harness protector from the stud bolt.

15. While pressing the wiring harness clip tabs in the direction of the arrows (1) shown in the figure, push out the wiring harness clip in the direction of the arrow (2) shown in the figure to detach the wiring harness clip tabs from the relay and fuse block lower cover No.1.



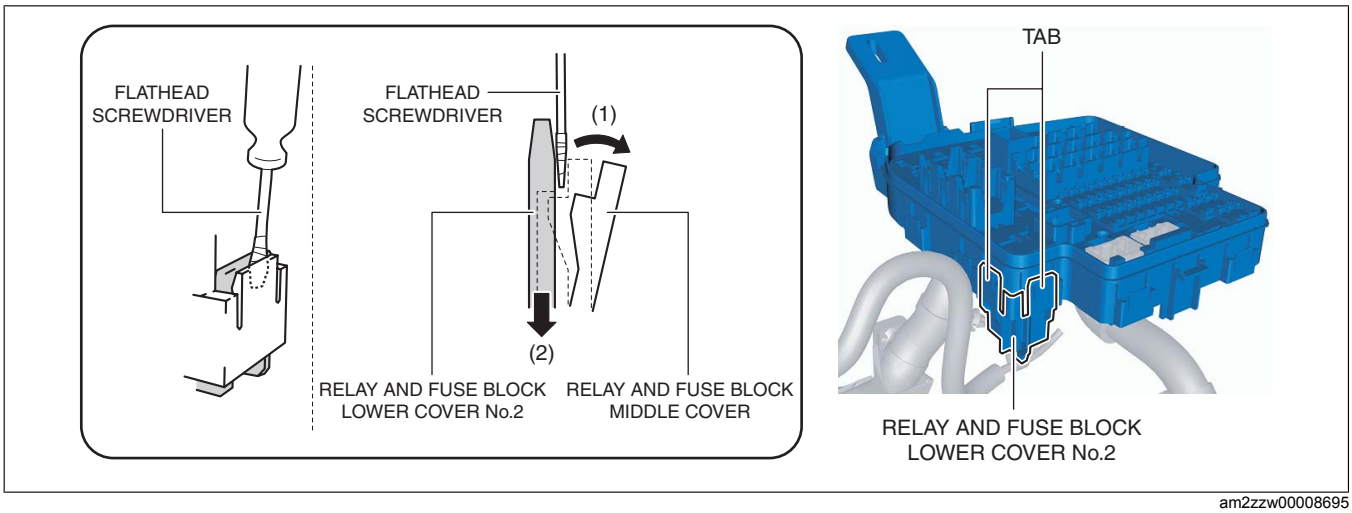
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16. Remove the wiring harness clips.
17. While pressing the relay and fuse block middle cover tab in the direction of the arrow (1) shown in the figure using a tape-wrapped flathead screwdriver, pull down the relay and fuse block lower cover No.1 in the direction of the arrow (2) to detach the relay and fuse block middle cover tab from the relay and fuse block lower cover No.1.



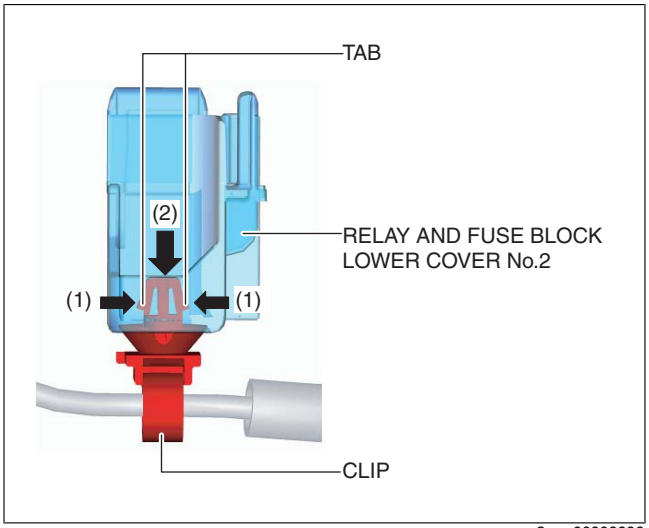
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18. Detach all of the relay and fuse block middle cover tabs and the relay and fuse block lower cover No.1.
19. Remove the relay and fuse block lower cover No.1.
20. While pressing the relay and fuse block middle cover tab in the direction of the arrow (1) shown in the figure using a tape-wrapped flathead screwdriver, pull down the relay and fuse block lower cover No.2 in the direction of the arrow (2) to detach the relay and fuse block middle cover tab from the relay and fuse block lower cover No.2.



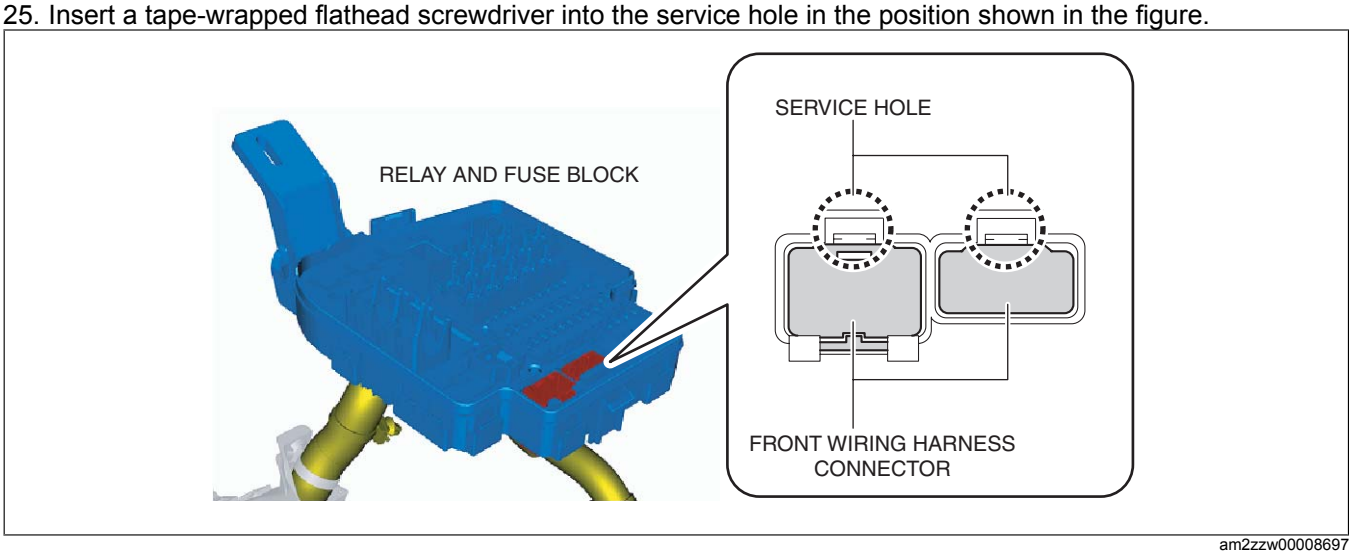
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21. Detach all of the relay and fuse block middle cover tabs and the relay and fuse block lower cover No.2.
22. Remove the relay and fuse block lower cover No.2.
23. While pressing the wiring harness clip tabs in the direction of the arrows (1) shown in the figure, push out the wiring harness clip in the direction of the arrow (2) shown in the figure to detach the wiring harness clip tabs from the relay and fuse block lower cover No.2.
24. Remove the wiring harness clip.



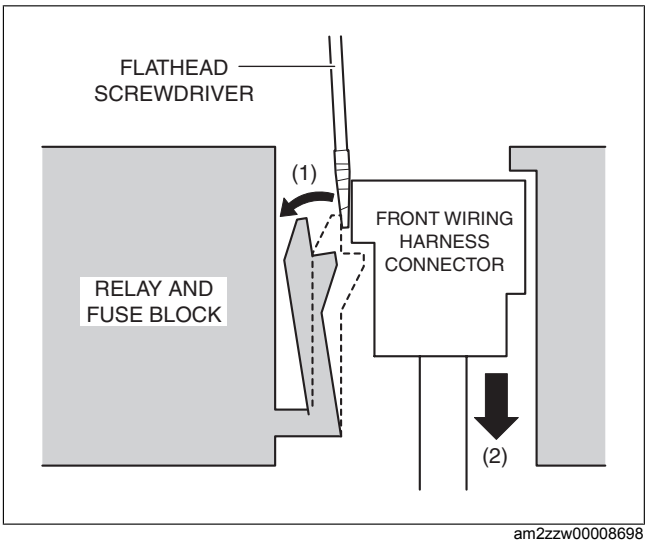
am2zzw00008696

25. Insert a tape-wrapped flathead screwdriver into the service hole in the position shown in the figure.



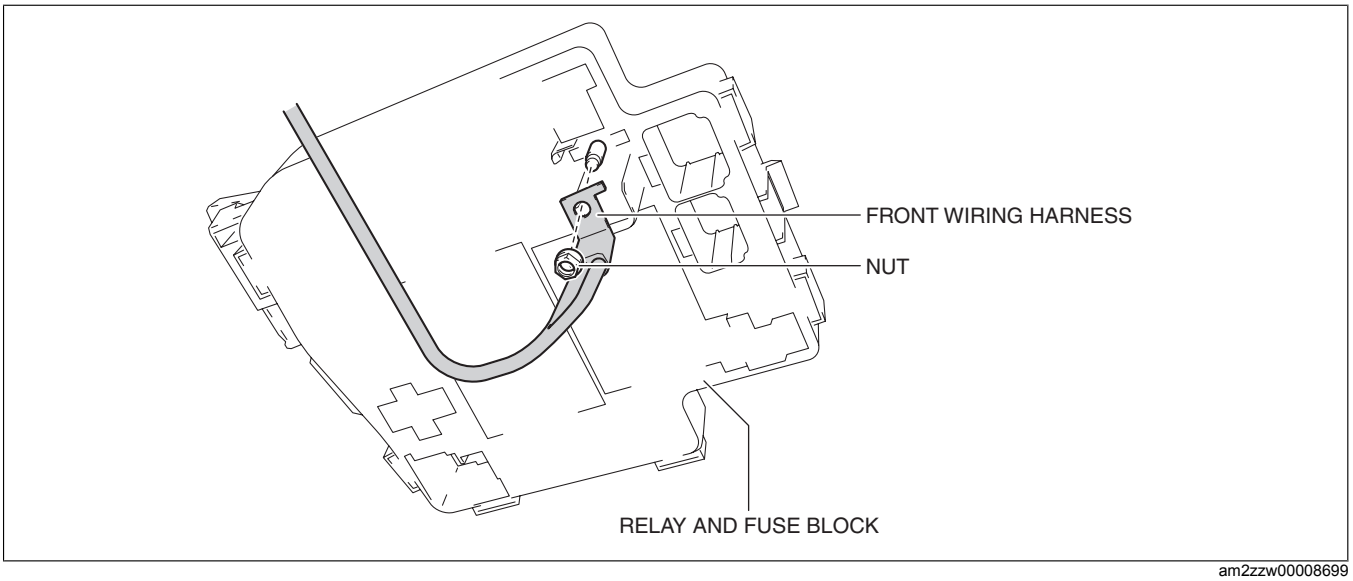
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26. While pressing the relay and fuse block tab in the direction of the arrow (1) shown in the figure using a tape-wrapped flathead screwdriver, pull down the front wiring harness connector in the direction of the arrow (2) to detach the relay and fuse block tab from the front wiring harness connector.
27. Pull out all the front wiring harness connectors from the relay and fuse block.



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28. Remove the nut.



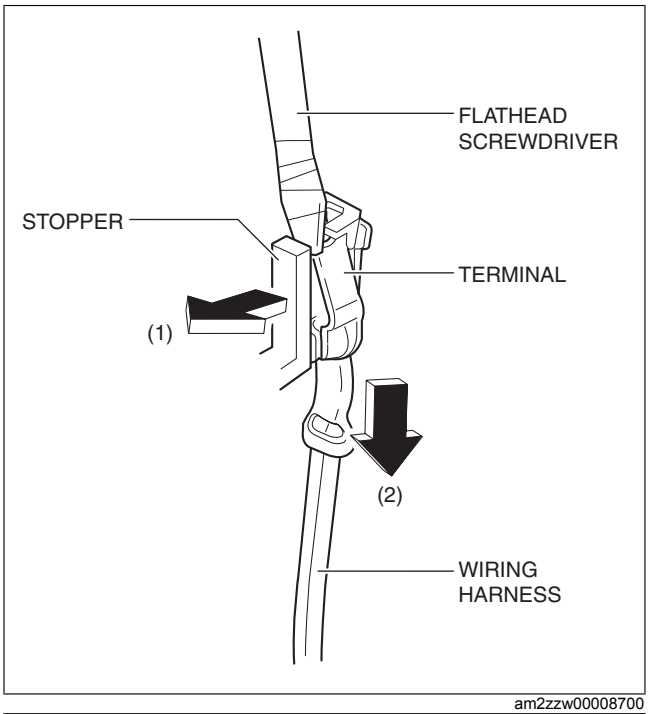
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29. Remove the front wiring harness.
30. Set the removed relay and fuse block aside so that it does not scratch the vehicle and parts.

Installation

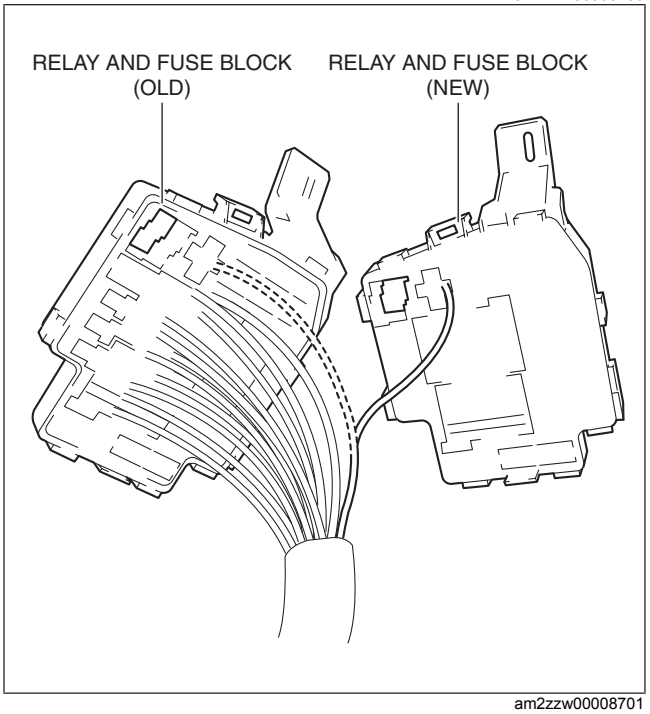
1. While pressing the stopper using a tape-wrapped flathead screwdriver in the direction of the arrow (1) shown in the figure, pull the wiring harness in the direction of the arrow (2) shown in the figure to detach the fuse terminal from the stopper.

   Caution
   - **If terminals are installed incorrectly, it could damage electrical accessories and burn wiring harnesses. When relocating terminals after replacing the relay and fuse block, line up the relay and fuse blocks of old and new parts and always verify each of the wiring harnesses in the wiring diagram so that the installation positions are correct.**
   - **A terminal disconnection could cause electronic components and the system to not operate normally due to poor contact. After terminal relocation, lightly pull the wiring harnesses to verify that they cannot be pulled out.**



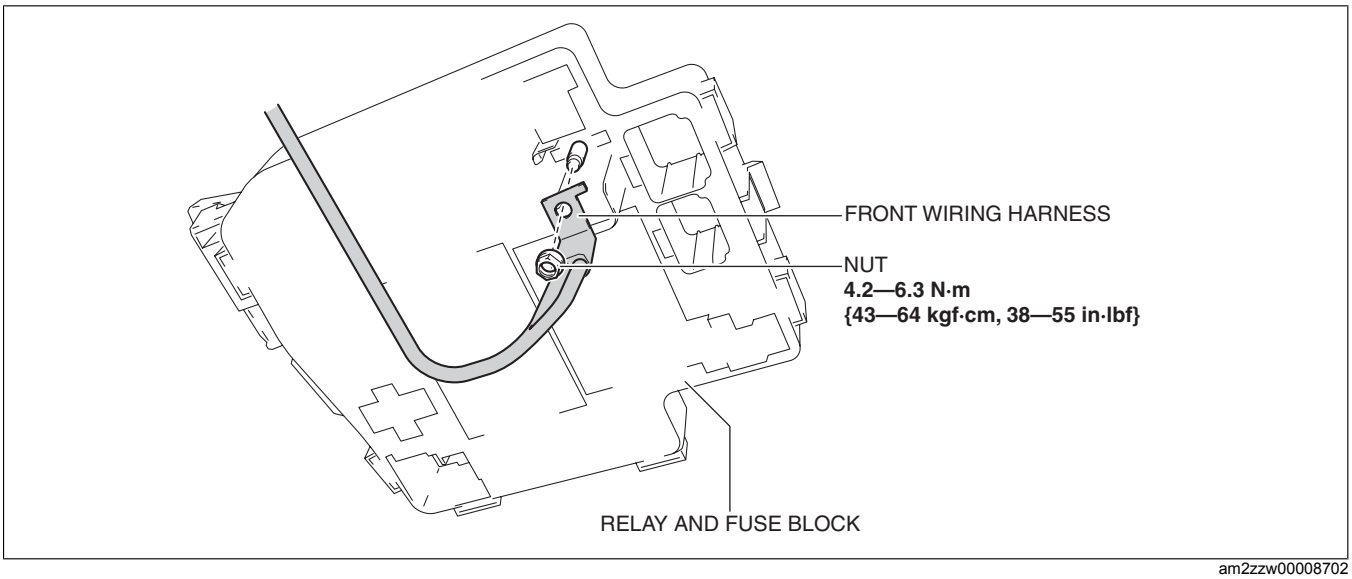
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2. Relocate all relay and fuse terminals to the new relay and fuse block.



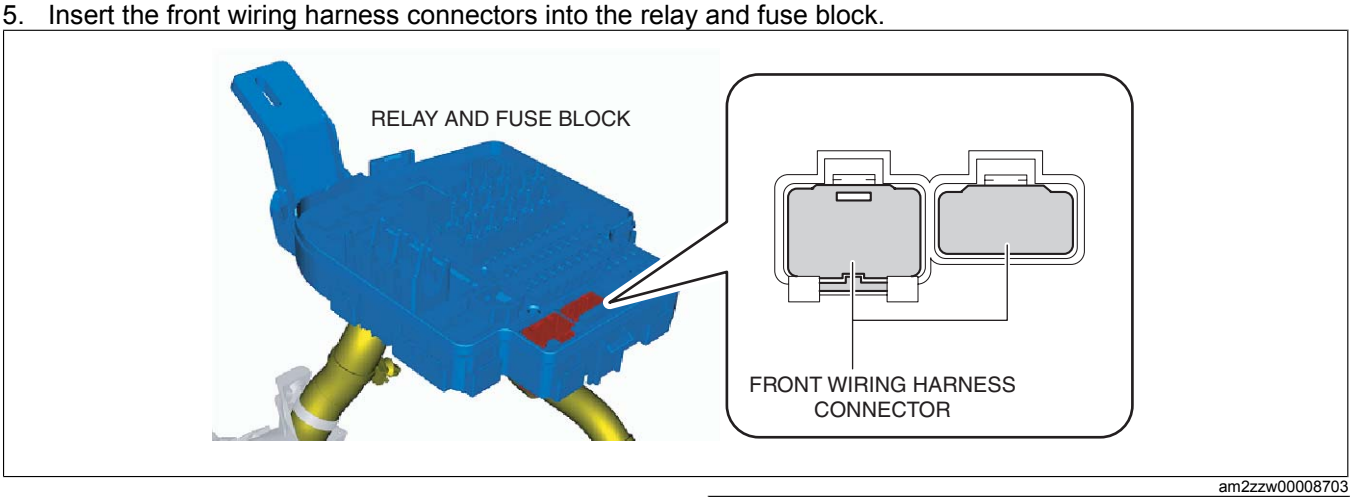
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3. Install the front wiring harness.



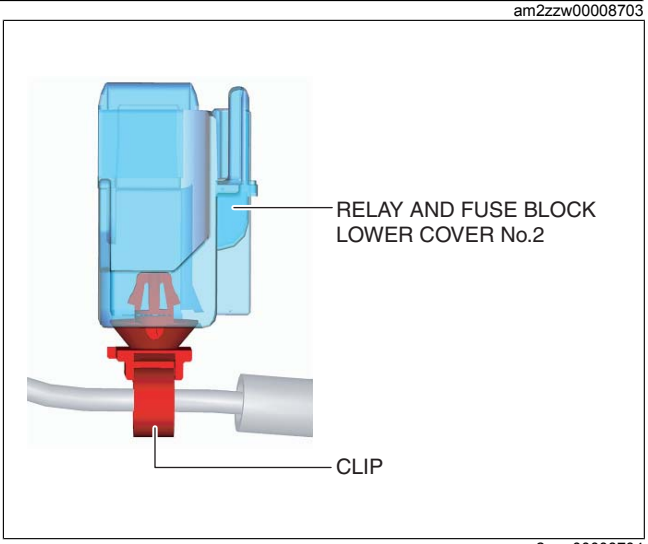
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4. Install the nut.
5. Insert the front wiring harness connectors into the relay and fuse block.



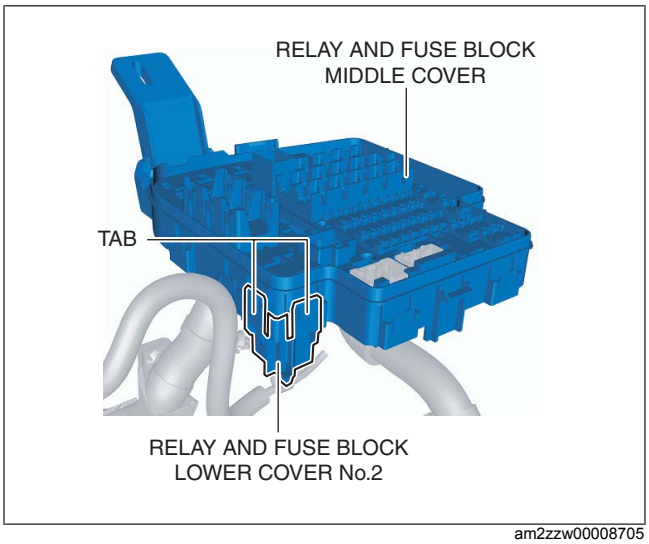
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6. Install the wiring harness clip to the relay and fuse block lower cover No.2.



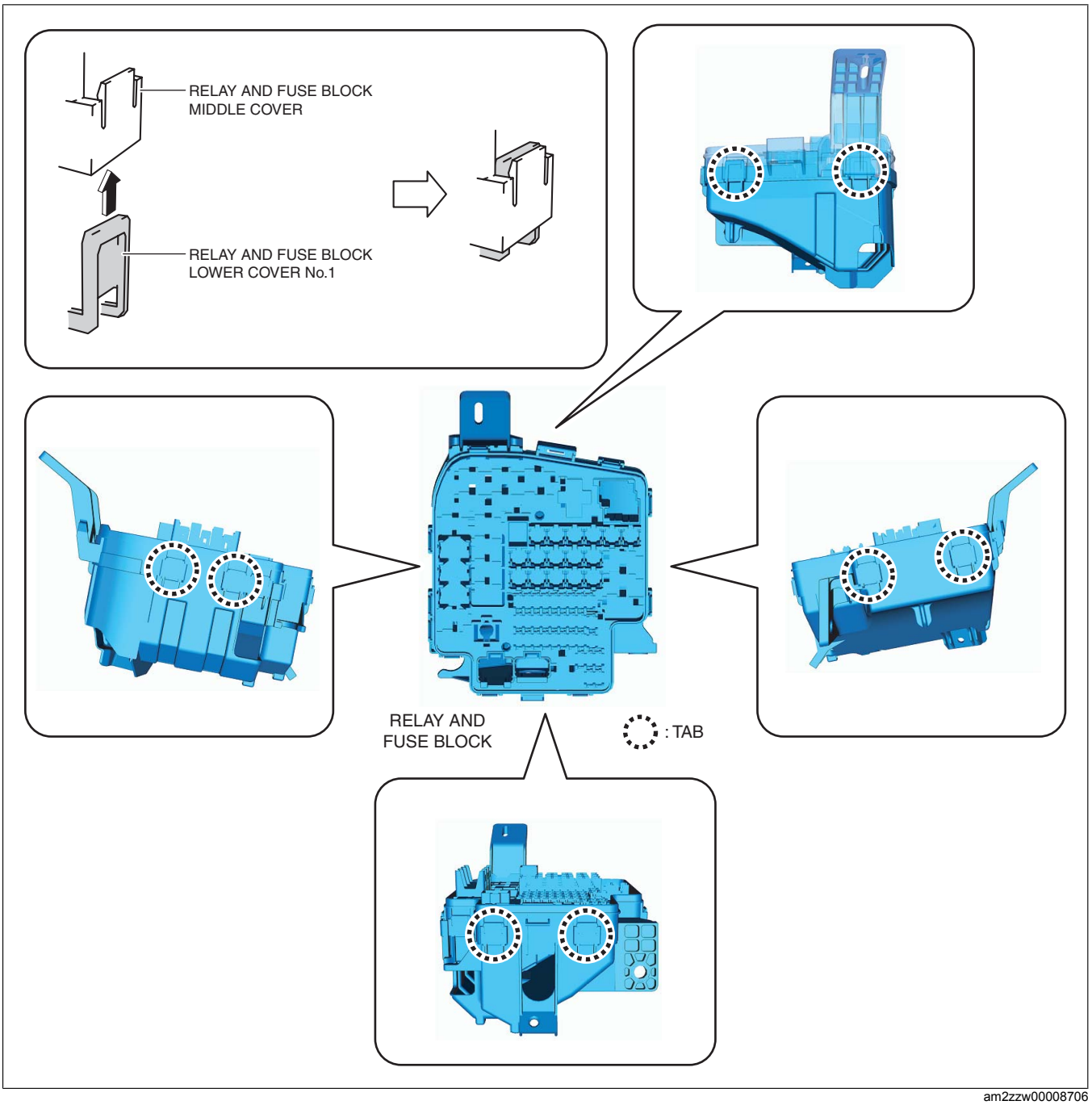
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7. Insert all of the relay and fuse block middle cover tabs into the relay and fuse block lower cover No.2.
8. Install the relay and fuse block lower cover No.2 to the relay and fuse block middle cover.



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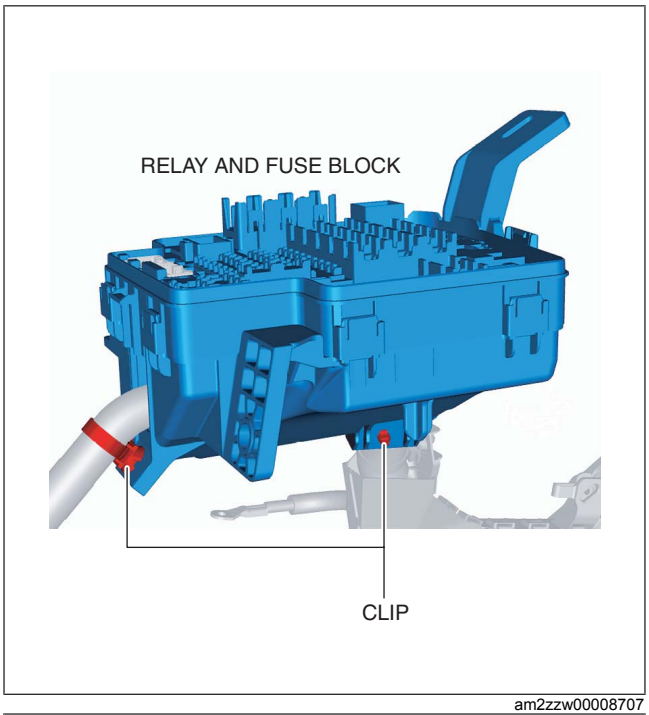
9. Insert all of the relay and fuse block middle cover tabs into the relay and fuse block lower cover No.1.



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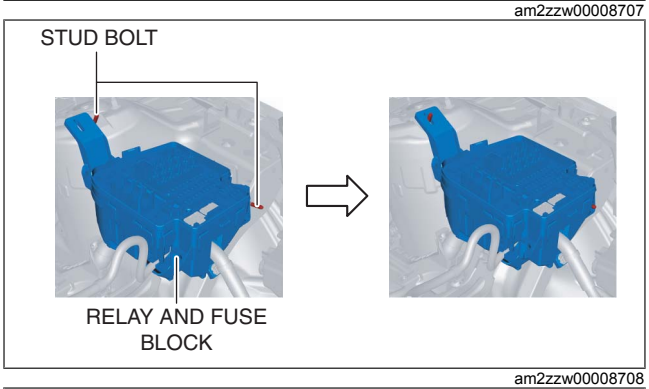
10. Install the relay and fuse block lower cover No.1 to the relay and fuse block middle cover.

11. Install the wiring harness clips.



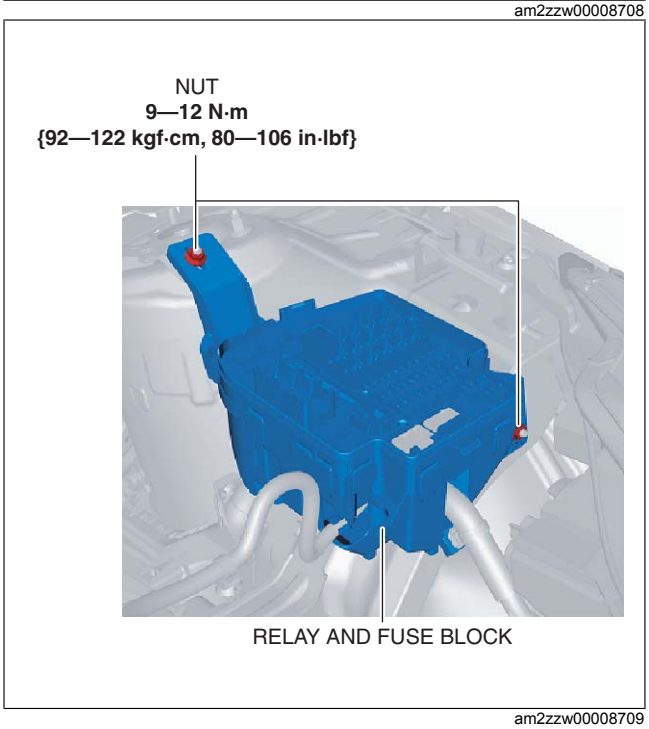
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12. Insert the relay and fuse block into the stud bolts.



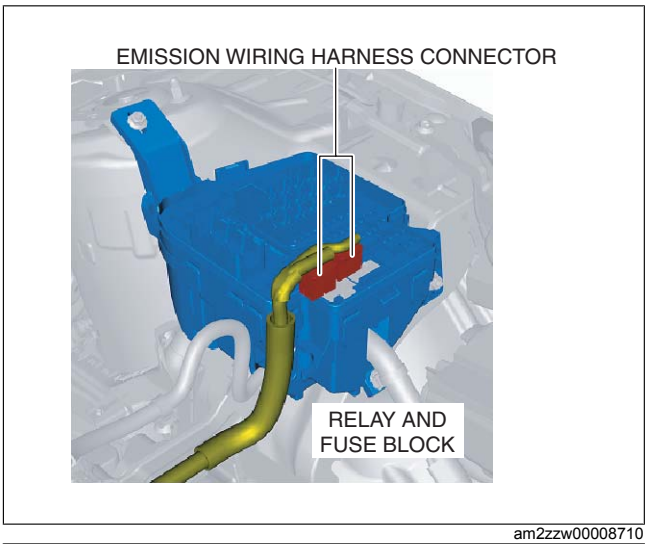
am2zzw00008708

13. Install the nuts.



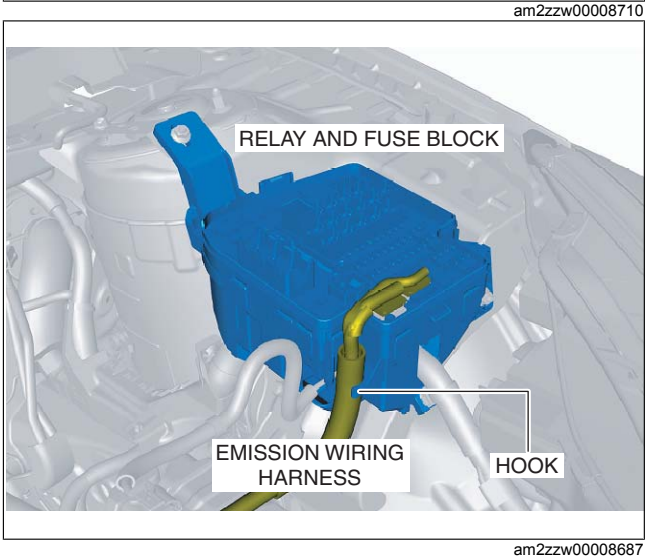
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14. Connect the emission wiring harness connectors.



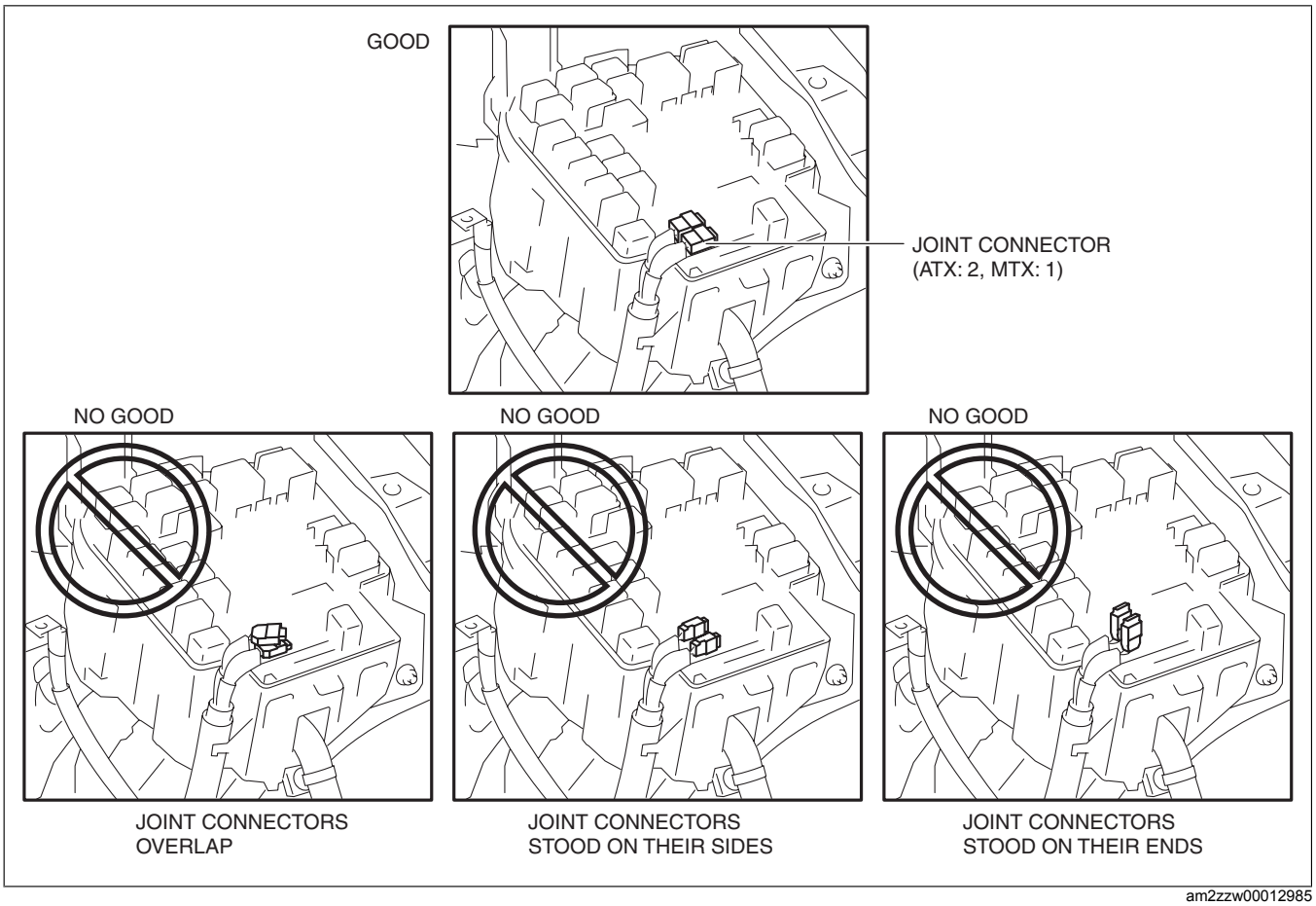
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15. Press the emission wiring harness into the relay and fuse block hook.
16. Install all the relays and fuses.



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17. For vehicles with joint connectors, verify that the joint connectors are in their normal positions.



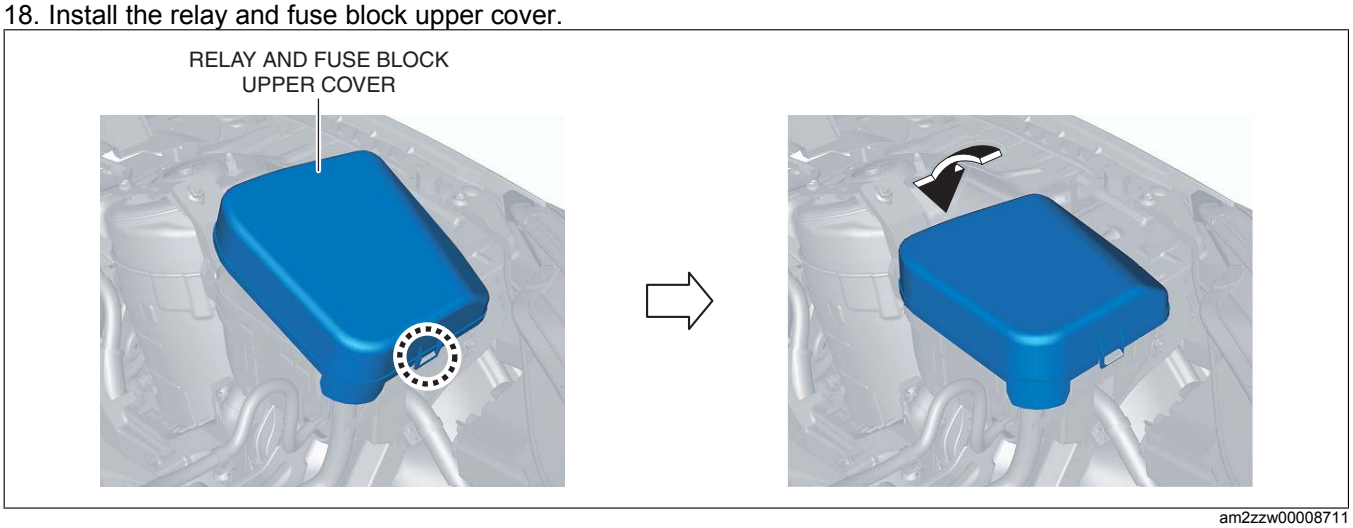
am2zzw00012985

- If the joint connectors are not in their normal positions, move them to their normal positions.

Caution

- **If the relay and fuse block upper cover is installed with the joint connectors not in their normal positions, damage to the relay and fuse block upper cover may occur. Before installing the relay and fuse block upper cover, always verify that the joint connectors are in their normal positions, and then install the relay and fuse block upper cover.**

18. Install the relay and fuse block upper cover.



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19. Install the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
20. Install the air cleaner assembly. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
21. Perform the "Verify Relay and Fuse Block Replacement". (See Verify relay and fuse block replacement.)

Verify relay and fuse block replacement

1. Temporarily connect the negative battery cable to verify that the related wiring harness is not heated.

2. Switch the ignition ON (engine off).
3. Verify that the fuse is not blown and the related wiring harness is not heated.
4. Perform the initial settings for the following systems:
   — i-stop control (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).)
   — Power window system (See POWER WINDOW SYSTEM INITIALIZATION PROCEDURE.)
   — Clock and audio
5. Verify DTCs using the **M-MDS**.
   • If a DTC is displayed, repair the malfunctioning part according to the corresponding DTC inspection.
     — After completion of repairs, clear all DTCs stored in the module.
6. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
7. Start the engine and maintain the idle status.
8. Operate all the switches of the interior/exterior lights of the vehicle and verify that all the lights illuminate and flash.
9. Operate the audio, climate control, wiper switch, and selector lever (ATX) and verify that they operate correctly.
10. Shift the selector lever to the P position, race the engine at approx. 3,000 rpm from engine idling, and verify that there is no malfunction. (ATX)
11. Perform a road test and verify that low speed, acceleration, shift-up, and shift-down conditions are normal. (ATX)